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The test consists of **2 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

	a	b	c	d	e	f	g
1.)	☐	☐	☐	☐	☐	☐	☐
2.)	☐	☐	☐	☐	☐	☐	☐

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) Let $L^{(1)} = \langle LC, Var, (\mathcal{P}(1) \cup \mathcal{P}(2) \cup \mathcal{F}(0) \cup \mathcal{F}(2)), Term, Form \rangle$ be a first-order language so that

$$Var = \{y_i \mid i \in \mathbb{N}\}, \quad \mathcal{P}(1) = \{Q\}, \quad \mathcal{P}(2) = \{R\}, \quad \mathcal{F}(0) = \{a\}, \quad \mathcal{F}(2) = \{g\}.$$

Which string of symbols are terms or formulas of the language $L^{(1)}$. Select the correct statements.

- (a) A term of the language $L^{(1)}$ is: Q
- (b) A term of the language $L^{(1)}$ is: $(y_3 = a)$
- 2

 (c) A term of the language $L^{(1)}$ is: $g(y_1, y_2)$
- (d) A formula of the language $L^{(1)}$ is: $\neg g(y_1, y_2)$
- (e) A formula of the language $L^{(1)}$ is: $\exists a R(a, a)$
- (f) A formula of the language $L^{(1)}$ is: $\forall y_1 \neg (Q(y_1) = a)$
- 2

 (g) A formula of the language $L^{(1)}$ is: $\forall y_1 \neg (Q(y_1) \vee (y_1 = a))$

2.) Let $L^{(1)}$ be a first-order language defined as follows:

$$L^{(1)} = \langle LC, \{x_1, x_2, \dots\}, \{P, Q, c\}, Term, Form \rangle$$

$$P \in \mathcal{P}(1), \quad Q \in \mathcal{P}(2), \quad c \in \mathcal{F}(0).$$

Let $\langle U, \varrho \rangle$ az $L^{(1)}$ be an interpretation of the language, so that:

- $U = \{u_1, u_2, u_3\}$
- $\varrho(P) = P' \quad P'(x) = \begin{cases} 1 & \text{if } x = u_1 \\ 0 & \text{otherwise} \end{cases}$
- $\varrho(Q) = Q' \quad Q'(x, y) = \begin{cases} 1 & \text{if } x = u_1 \text{ and } y = u_1 \\ 0 & \text{otherwise} \end{cases}$
- $\varrho(c) = u_2$

Let v be an assignment relying on the interpretation, so that $v : Var \rightarrow U$ and

$$v(x_1) = u_3, \quad v(x_2) = u_1$$

Which of the formulas below are true?

(a) $| (P(x_2) \supset Q(x_1, x_2)) |_v^{(U, \varrho)}$

(b) $| \neg Q(x_2, x_2) |_v^{(U, \varrho)}$

2 $| (Q(x_1, x_2) \supset P(x_1)) |_v^{(U, \varrho)}$

2 $| \neg (P(x_1) \vee P(c)) |_v^{(U, \varrho)}$

2 $| (Q(c, c) \equiv P(x_1)) |_v^{(U, \varrho)}$

(f) $| (Q(c, c) \equiv P(x_2)) |_v^{(U, \varrho)}$

(g) $| ((c = c) \supset (x_2 = x_1)) |_v^{(U, \varrho)}$